z-Scores & Linear Transformations of Data

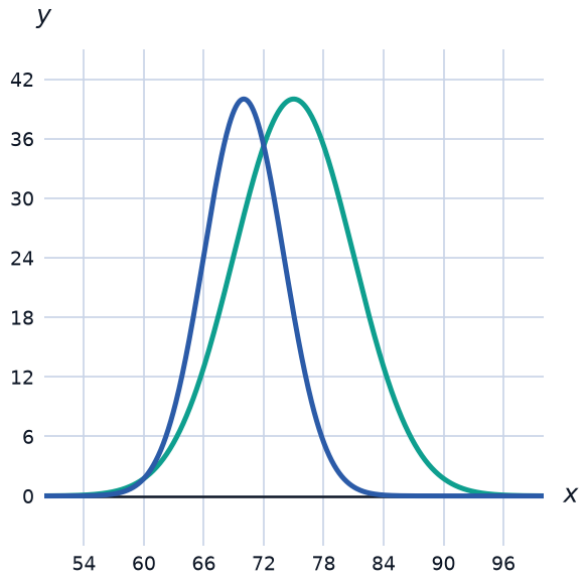
Computing z-scores

1 A test has mean 70 and standard deviation 5. Find the z-score for a student who scored 82.

2 Explain in words what a z-score of -1.5 means, in the context of a data value's position relative to its distribution.

Comparing values across different distributions

3 On Test A (mean 75, standard deviation 6), a student scores 87. On Test B (mean 70, standard deviation 4), a different student scores 77. Compare the two students' relative performance using z-scores.



Effect of adding a constant to every data value

4 A data set has mean 50 and standard deviation 8. If 10 is added to every value, find the new mean and new standard deviation.

5 Explain, without doing a full calculation, why adding a constant to every data value does not change the standard deviation.

Effect of multiplying every data value by a constant

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

6 A data set has mean 20 and standard deviation 3. If every value is multiplied by 4, find the new mean and new standard deviation.

7 A distance measured in miles has mean 10 and standard deviation 2. Convert both to kilometers (multiply by 1.609), and find the new mean and standard deviation, to two decimal places.

Visualizing a standardized (z-score) distribution

8 The graph shows the standard normal distribution, which has mean 0 and standard deviation 1 — the distribution that results after standardizing any normal data set using z-scores.



a State the mean and standard deviation shown by this curve.

b Explain why standardizing a data set using $z = \dfrac{x - \mu}{\sigma}$ always produces a distribution with mean 0 and standard deviation 1, regardless of the original data's mean and standard deviation.

Applying linear transformations in context

9 A factory's daily output (in units) has mean 500 and standard deviation 30. Management adds a fixed bonus batch of 50 units to every day's count, then reports output in dozens (divide by 12). Find the new mean and standard deviation, to two decimal places.

Finding a raw value from a z-score

10 A distribution has mean $\mu = 64$ and standard deviation $\sigma = 5$. Find the data value corresponding to a z-score of $z = -1.8$.

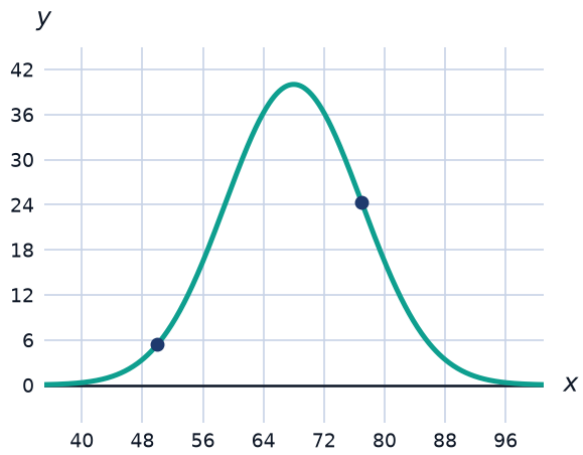
11 A distribution has mean $\mu = 200$ and standard deviation $\sigma = 15$. If a data value has $z = 2.2$, find the data value.

Combined shift-and-scale transformations

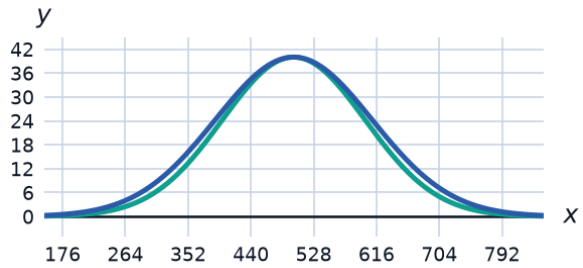
© ClipSAT: By Mr. Mohamed Abdallah +966597688647

12 A teacher curves test scores (mean 65, standard deviation 10) by first adding 5 points to every score, then multiplying the result by 1.1. Find the new mean and new standard deviation, to two decimal places.

13 A weight distribution has mean $\mu = 68$ kg and standard deviation $\sigma = 9$ kg. Find the z-scores for weights of 77 kg and 50 kg.



14 A national math exam has mean 500 and standard deviation 100, while a national verbal exam has mean 500 and standard deviation 110. A student scores 640 on the math exam and 625 on the verbal exam. Even though the verbal raw score is lower, use z-scores to determine which exam the student performed better on relative to other test-takers, and explain your reasoning.



15 A shipping company records package weights with mean 18 kg and standard deviation 2.5 kg. The company then converts every measurement from kilograms to pounds by multiplying by 2.20462, and separately subtracts 1 kg from every package to account for packaging weight before converting. Describe, step by step, how each operation (the subtraction, then the multiplication) affects the mean and standard deviation.

Multiple choice — easy 15

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

16 A distribution has mean $\mu = 51$ and standard deviation $\sigma = 25$. Find the z-score for a value of $x = 41$.

a. -0.40

b. 0.40

c. -0.90

d. 0.10

17 A distribution has mean $\mu = 80$ and standard deviation $\sigma = 4$. Find the z-score for a value of $x = 126$.

a. -11.50

b. 11.00

c. 11.50

d. 12.00

18 A distribution has mean $\mu = 30$ and standard deviation $\sigma = 2$. Find the z-score for a value of $x = 66$.

a. 18.50

b. -18.00

c. 18.00

d. 17.50

19 A distribution has mean $\mu = 94$ and standard deviation $\sigma = 5$. Find the z-score for a value of $x = 71$.

a. 4.60

b. -5.10

c. -4.60

d. -4.10

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

20 A distribution has mean $\mu = 50$ and standard deviation $\sigma = 2$. Find the z-score for a value of $x = -5$.

a. -27.50

b. -28.00

c. -27.00

d. 27.50

21 A distribution has mean $\mu = 85$ and standard deviation $\sigma = 2$. Find the z-score for a value of $x = 133$.

a. 24.50

b. 24.00

c. -24.00

d. 23.50

22 A distribution has mean $\mu = 77$ and standard deviation $\sigma = 25$. Find the z-score for a value of $x = 50$.

a. -0.58

b. 1.08

c. -1.58

d. -1.08

23 A distribution has mean $\mu = 89$ and standard deviation $\sigma = 8$. Find the z-score for a value of $x = 40$.

a. -6.12

b. -6.62

c. 6.12

d. -5.62

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

24 A distribution has mean $\mu = 98$ and standard deviation $\sigma = 5$. Find the z-score for a value of $x = 154$.

a. 10.70

b. 11.70

c. 11.20

d. -11.20

25 A distribution has mean $\mu = 78$ and standard deviation $\sigma = 8$. Find the z-score for a value of $x = 19$.

a. 7.38

b. -7.38

c. -7.88

d. -6.88

26 A distribution has mean $\mu = 79$ and standard deviation $\sigma = 25$. Find the z-score for a value of $x = 30$.

a. -1.46

b. -1.96

c. -2.46

d. 1.96

27 A distribution has mean $\mu = 80$ and standard deviation $\sigma = 20$. Find the z-score for a value of $x = 119$.

a. 1.95

b. 1.45

c. 2.45

d. -1.95

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

28 A distribution has mean $\mu = 59$ and standard deviation $\sigma = 4$. Find the z-score for a value of $x = 63$.

a. 1.50

b. -1.00

c. 0.50

d. 1.00

29 A distribution has mean $\mu = 69$ and standard deviation $\sigma = 2$. Find the z-score for a value of $x = 44$.

a. -12.00

b. 12.50

c. -12.50

d. -13.00

30 A distribution has mean $\mu = 51$ and standard deviation $\sigma = 4$. Find the z-score for a value of $x = 50$.

a. -0.75

b. -0.55

c. -0.25

d. 0.25

Multiple choice — medium 15

31 A distribution has mean $\mu = 85$ and standard deviation $\sigma = 25$. Find the data value corresponding to a z-score of $z = -1.4$.

a. 49

b. 51

c. 120

d. 50

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

32 A distribution has mean $\mu = 168$ and standard deviation $\sigma = 10$. Find the data value corresponding to a z-score of $z = -2.8$.

a. 139

b. 196

c. 141

d. 140

33 A distribution has mean $\mu = 49$ and standard deviation $\sigma = 5$. Find the data value corresponding to a z-score of $z = -1.2$.

a. 42

b. 55

c. 43

d. 44

34 A distribution has mean $\mu = 76$ and standard deviation $\sigma = 20$. Find the data value corresponding to a z-score of $z = 1.3$.

a. 101

b. 103

c. 102

d. 50

35 A distribution has mean $\mu = 130$ and standard deviation $\sigma = 10$. Find the data value corresponding to a z-score of $z = -0.4$.

a. 134

b. 126

c. 125

d. 127

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

36 A distribution has mean $\mu = 180$ and standard deviation $\sigma = 20$. Find the data value corresponding to a z-score of $z = -2.2$.

a. 224

b. 137

c. 135

d. 136

37 A distribution has mean $\mu = 101$ and standard deviation $\sigma = 20$. Find the data value corresponding to a z-score of $z = -2.9$.

a. 44

b. 159

c. 43

d. 42

38 A distribution has mean $\mu = 123$ and standard deviation $\sigma = 25$. Find the data value corresponding to a z-score of $z = 0.4$.

a. 113

b. 132

c. 133

d. 134

39 A distribution has mean $\mu = 132$ and standard deviation $\sigma = 4$. Find the data value corresponding to a z-score of $z = -2.5$.

a. 142

b. 122

c. 123

d. 121

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

40 A distribution has mean $\mu = 189$ and standard deviation $\sigma = 5$. Find the data value corresponding to a z -score of $z = 0.8$.

a. 193

b. 194

c. 185

d. 192

41 A distribution has mean $\mu = 50$ and standard deviation $\sigma = 20$. Find the data value corresponding to a z -score of $z = 2.4$.

a. 98

b. 97

c. 99

d. 2

42 A distribution has mean $\mu = 144$ and standard deviation $\sigma = 20$. Find the data value corresponding to a z -score of $z = 2.1$.

a. 185

b. 102

c. 186

d. 187

43 A distribution has mean $\mu = 57$ and standard deviation $\sigma = 10$. Find the data value corresponding to a z -score of $z = -1.9$.

a. 37

b. 38

c. 39

d. 76

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

44 A distribution has mean $\mu = 143$ and standard deviation $\sigma = 20$. Find the data value corresponding to a z-score of $z = 0.8$.

a. 127

b. 158

c. 160

d. 159

45 A distribution has mean $\mu = 111$ and standard deviation $\sigma = 8$. Find the data value corresponding to a z-score of $z = 0.5$.

a. 115

b. 107

c. 114

d. 116

Multiple choice — hard 15

46 On Test A (mean 79, standard deviation 4), a student scores 94. On Test B (mean 88, standard deviation 8), a different student scores 99. Using z-scores, which student performed better relative to their own test's distribution?

a. Test A

b. Test B

c. Cannot be determined without more information

d. Both are exactly the same

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

47 On Test A (mean 52, standard deviation 4), a student scores 53. On Test B (mean 54, standard deviation 5), a different student scores 61. Using z-scores, which student performed better relative to their own test's distribution?

a. Test A

b. Both are exactly the same

c. Cannot be determined without more information

d. Test B

48 On Test A (mean 67, standard deviation 6), a student scores 69. On Test B (mean 55, standard deviation 5), a different student scores 64. Using z-scores, which student performed better relative to their own test's distribution?

a. Test B

b. Both are exactly the same

c. Cannot be determined without more information

d. Test A

49 On Test A (mean 56, standard deviation 6), a student scores 76. On Test B (mean 67, standard deviation 4), a different student scores 72. Using z-scores, which student performed better relative to their own test's distribution?

a. Both are exactly the same

b. Cannot be determined without more information

c. Test B

d. Test A

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

50 On Test A (mean 90, standard deviation 10), a student scores 94. On Test B (mean 76, standard deviation 4), a different student scores 86. Using z-scores, which student performed better relative to their own test's distribution?

a. Both are exactly the same

b. Test A

c. Test B

d. Cannot be determined without more information

51 A data set has mean 75 and standard deviation 17. First, -4 is added to every value; then every value is multiplied by 4. Find the new mean and standard deviation as an ordered pair (mean, SD).

a. $(284, 17)$

b. $(285, 68)$

c. $(284, 68)$

d. $(300, 68)$

52 A data set has mean 40 and standard deviation 18. First, 5 is added to every value; then every value is multiplied by 2. Find the new mean and standard deviation as an ordered pair (mean, SD).

a. $(90, 18)$

b. $(80, 36)$

c. $(91, 36)$

d. $(90, 36)$

53 A data set has mean 75 and standard deviation 18. First, 2 is added to every value; then every value is multiplied by 5. Find the new mean and standard deviation as an ordered pair (mean, SD).

a. $(375, 90)$

b. $(386, 90)$

c. $(385, 90)$

d. $(385, 18)$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

54 A data set has mean 40 and standard deviation 3. First, -7 is added to every value; then every value is multiplied by 4. Find the new mean and standard deviation as an ordered pair (mean, SD).

a. $(132, 12)$

b. $(133, 12)$

c. $(160, 12)$

d. $(132, 3)$

55 A data set has mean 61 and standard deviation 17. First, 3 is added to every value; then every value is multiplied by 5. Find the new mean and standard deviation as an ordered pair (mean, SD).

a. $(305, 85)$

b. $(320, 85)$

c. $(321, 85)$

d. $(320, 17)$

56 A distribution has mean $\mu = 97$ and standard deviation $\sigma = 4$. Find the z-scores for values of 113 and 84, as an ordered pair (z_{113}, z_{84}), to two decimal places.

a. $(4.30, -3.25)$

b. $(4.00, -3.25)$

c. $(4.00, -3.55)$

d. $(-3.25, 4.00)$

57 A distribution has mean $\mu = 41$ and standard deviation $\sigma = 8$. Find the z-scores for values of 52 and 31, as an ordered pair (z_{52}, z_{31}), to two decimal places.

a. $(1.38, -1.25)$

b. $(1.68, -1.25)$

c. $(-1.25, 1.38)$

d. $(1.38, -1.55)$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

58 A distribution has mean $\mu = 82$ and standard deviation $\sigma = 8$. Find the z-scores for values of 104 and 70, as an ordered pair (z_{104}, z_{70}), to two decimal places.

a. $(2.75, -1.50)$

b. $(-1.50, 2.75)$

c. $(2.75, -1.80)$

d. $(3.05, -1.50)$

59 A distribution has mean $\mu = 53$ and standard deviation $\sigma = 5$. Find the z-scores for values of 74 and 47, as an ordered pair (z_{74}, z_{47}), to two decimal places.

a. $(4.50, -1.20)$

b. $(4.20, -1.50)$

c. $(-1.20, 4.20)$

d. $(4.20, -1.20)$

60 A distribution has mean $\mu = 50$ and standard deviation $\sigma = 3$. Find the z-scores for values of 57 and 43, as an ordered pair (z_{57}, z_{43}), to two decimal places.

a. $(2.33, -2.63)$

b. $(2.33, -2.33)$

c. $(-2.33, 2.33)$

d. $(2.63, -2.33)$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647